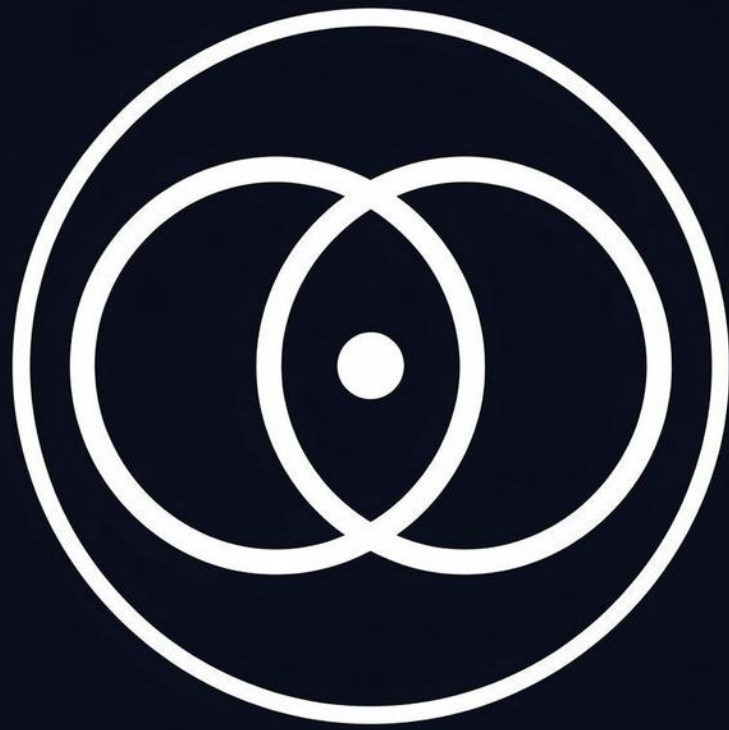




P22

**The Memory Wall Is Not the
Coordination Wall: A
Capacity–Quality–Coordination
Tripartition behind
Warp-Cortex's Architectural
Rescaling**

内存墙不等于协调墙：规模二分

The Memory Wall Is Not the Coordination Wall: A Capacity–Quality–Coordination Tripartition behind Warp-Cortex's Architectural Rescaling

Working Title (EN): The Memory Wall Is Not the Coordination Wall: A Capacity–Quality–Coordination Tripartition behind Warp-Cortex's Architectural Rescaling

Volume P22 • sister paper to P9 "The Quality Saturation Law" and P21 "The MAPF Extrapolation Boundary"
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One sentence first: Warp-Cortex tore down the memory wall of "running a hundred Agents on one consumer-grade GPU"—a real contribution; but the Singleton Weight Sharing it uses (all Agents sharing one copy of weights) is essentially **forced homogenization**, preserving intact the "homogeneous-copy quality saturation" P9 has proved. Memory can scale $\neq$ capability can scale $\neq$ coordination can scale; these are three things, and treating them as one is the most common equivocation of the "million-Agent narrative."

Evidence caliber (stated only once in the full text): this paper is a theory/architecture-analysis paper containing no new UDOS measured numbers; the UDOS conclusions cited all return to v7.5.0 P9/P1 and their reports7/*.json (cpu-proto / verified grading). External literature is verified per EVIDENCE_LEDGER_v76.md: Warp-Cortex's (arXiv:2601.01298) Singleton Weight Sharing, Topological Synapse, Referential Injection are true; "single RTX 4090, 100 concurrent Agents, 2.2 GB VRAM, theoretical capacity 1,000+ Agents" is **paper-reported**, "million" is the paper's self-stated **theoretical extrapolation** and explicitly before compute latency becomes the bottleneck—**actually measured only about the hundred level, not the million level**, and this paper cites it at this caliber.

Abstract

Warp-Cortex (arXiv:2601.01298) is the most-cited architectural breakthrough in the 2026 "million-Agent cognitive scaling" narrative. This paper does two things. First, **accurately restate which wall it really tore down:** using Singleton Weight Sharing (all Agents sharing the same model instance's weights) + Topological Synapse (inspired by topological-data-analysis landmarking, treating KV-cache as a latent point cloud and sparsifying to retain persistent-homology features) it presses memory complexity from $\mathcal{O}(N \cdot L)$ to $\mathcal{O}(1)$ weights $\mathcal{O}(N \cdot k)$ context; the paper reports one RTX 4090 actually measuring about 100 concurrent Agents, 2.2 GB VRAM. Second, **point out which wall it did not tear down:** Singleton Weight Sharing makes all Agents share one copy of parameters, which at the architecture layer is what P9 calls "replication of the same near-correlated source"—they naturally carry no independent information. From this the "scale tripartition" is proposed: memory capacity (Warp-Cortex solves), coordination overhead (P1 solves), quality diversity (P9/P20 solve) are three independent failure boundaries, and tearing down one does not equal tearing down all. This paper simultaneously clarifies the title's "million Agents" is theoretical extrapolation, actually measured only the hundred level, and gives two falsifiable conditions.

Keywords: memory wall; Singleton Weight Sharing; Topological Synapse; KV-cache; homogenization; scale bipartition; capacity $\neq$ quality $\neq$ coordination

Structured Abstract (Background problem → Argument → Evidence → Contributions)

- **Background problem:** the "million-scale Agent" narrative conflates memory, coordination, and quality, three different bottlenecks. An architecture paper claims "consumer hardware runs million Agents," and readers naturally assume "million Agents are stronger," but the paper often solves only one of the walls.
 - **Core argument:** Warp-Cortex tears down the **memory wall** ($O(N \cdot L) \rightarrow O(1) + O(N \cdot k)$), not the **quality wall**; its Singleton Weight Sharing forces homogenization, directly conflicting with P9's homogenization saturation law, so "memory can scale" cannot imply "capability can scale."
 - **Evidence threads:** (1) Warp-Cortex's two mechanisms (paper-reported); (2) measured 100 Agents/2.2GB, theoretical thousand level, "million" an extrapolation; (3) P9 homogeneous copies $mse(k) = a + b/k$, $b \rightarrow 0$ (cpu-proto); (4) P1 coordination fan-in $O(N) \rightarrow O(\log N)$ (cpu-proto).
 - **Contributions:** splitting "scale" into memory capacity/coordination overhead/quality diversity three independent dimensions; pointing out singleton weight sharing is the architecture-layer source of homogenization; clarifying "million" as extrapolation; two falsifiable conditions.
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1 Introduction: One Wall Falling Does Not Equal All Walls Falling

In 2026, the bottleneck narrative of multi-Agent systems went through several switches. First "single-Agent capability insufficient," so several were opened in parallel; then "too many in parallel don't fit memory," so someone optimized memory; then in promotion it quietly became "the million-Agent era is here." Hidden in this jump is a logical hole: **memory fitting, and after fitting these Agents coordinating into higher quality, are two things.**

Warp-Cortex is the most worth dissecting sample in this hole. Its architectural contribution is real and elegant: letting one consumer-grade GPU simultaneously hold the cognitive processes of a hundred concurrent Agents. But what this paper does is separate "what it tore down" and "what it did not tear down"—because mixing these two into one directly misleads UDOS's architecture decisions.

2 Which Wall Warp-Cortex Tears Down

2.1 Two Mechanisms

Verified per EVIDENCE_LEDGER_v76.md (2026-09-21), the core of Warp-Cortex (Warp-Cortex: An Asynchronous, Memory-Efficient Architecture for Million-Agent Cognitive Scaling on Consumer Hardware, arXiv:2601.01298) is two mechanisms:

- **Singleton Weight Sharing:** all Agents share the same model instance's weights, not copying a set of parameters per Agent.
- **Topological Synapse:** treating each Agent's KV-cache as a point cloud in latent space, borrowing topological-data-analysis (TDA) landmarking to sparsify, retaining persistent-homology features (i.e. retaining "shape" rather than all points).

Together they press memory complexity from the traditional $O(N \cdot L)$ of each Agent storing its own weights+context to **weight part $O(1)$** (one shared copy) + **context part $O(N \cdot k)$** (each Agent storing only a sparse KV-cache subset, k far smaller than L). This is a real memory-wall

breakthrough. The paper also introduces Referential Injection, allowing asynchronous sub-Agents to influence the main generation without interrupting the main stream.

2.2 Honest Distinction of Measurement and Extrapolation

The paper reports: one RTX 4090 actually measuring about 100 concurrent Agents, 2.2 GB VRAM, theoretical capacity over 1,000 Agents. But the title's "million" is **theoretical extrapolation**, and the paper explicitly notes this is "before compute latency becomes the bottleneck." That is, measurement is the hundred level, the thousand level is theoretical calculation, the million level is title narrative. This volume follows P1's explicit/analytical discipline: measured 100 is explicit, thousand level analytical, million a title claim.

Figure 1 Memory complexity $O(N \cdot L) \rightarrow O(1) + O(N \cdot k)$

Figure 1 Left: traditional per-Agent weight storage, memory $O(N \cdot L)$; right: Warp-Cortex singleton weights $O(1)$ + per-Agent sparse KV-cache $O(N \cdot k)$. Paper reports 100 Agents/2.2GB measured, thousand level theoretical extrapolation. Conceptual illustration, not measured coordinates.

3 Which Wall It Did Not Tear Down: Singleton Weight Sharing Is Forced Homogenization

3.1 Shared Weights = Same-Source Replication

This is this paper's core argument. Singleton Weight Sharing makes all Agents share one copy of model weights. What does this mean? It means these hundred Agents are **same-source** on "where knowledge comes from"—they see the same parameters. Their differences lie only in KV-cache (i.e. respective context), not in the weights themselves.

Putting this structure into P20/P9's language: these hundred Agents are **copies of the same near-correlated source**. P9 in the cpu-proto experiment deliberately constructed homogenized copies of $\text{corr} \approx 0.9999$ and found adding 15 more copies does not lower MSE ($a=0.009252$, $b=-4.23 \times 10^{-5}$). Warp-Cortex's Singleton Weight Sharing at the architecture layer **defaults** to this homogenization—you need not "deliberately construct," it is naturally so.

3.2 Memory Wall vs Quality Wall

So Warp-Cortex tears down the **memory wall** (whether it can fit), not the **quality wall** (whether quality rises after fitting). Its value is to lower the memory cost of enlarging N ; it adds no independent information source (D unchanged). In P20's bipartition: it lets N reach 1000 but D remains about 1.

This is not a criticism of Warp-Cortex—its paper title and motivation honestly fall on "memory efficient." This is a warning to **those citing it**: do not read "memory can scale" as "capability can scale."

3.3 The Three-Wall Division

Generalizing this logic, the "scale" of multi-Agent systems in fact has three independent walls:

Table 1 Three independent walls: who tore down which

Wall	Question asked	Who tore it down (per v7.6.0 caliber)	After tearing down
Memory wall	Does memory fit N Agents?	Warp-Cortex (paper-reported)	N can reach hundreds/theoretical thousands
Coordination wall	Do messages between Agents explode?	UDOS P1 (cpu-proto)	Fan-in $O(N)$ to $O(\log N)$
Quality wall	Does adding Agents raise quality?	UDOS P9/P20 (cpu-proto)	Homogenization $\rightarrow 0$, needs independent sources

The three walls' solutions are not substitutable: Warp-Cortex makes N feasible in memory, P1 makes N feasible in coordination, P9 tells you even after the first two walls are torn down, quality may still be pinned at the saturation floor α .

Figure 2 Three independent walls

Figure 2 The memory wall (Warp-Cortex), coordination wall (P1), quality wall (P9/P20) are three independent failure boundaries; tearing down one does not equal tearing down all. Conceptual illustration.

4 Frontal Rendezvous with P9: The Bipartition of Capacity Scaling and Quality Scaling

4.1 Warp-Cortex's Agents Naturally Carry No Independent Information

P9's homogenized-copy construction ($\text{corr} \approx 0.9999$) is "deliberately simulating expansion"; Warp-Cortex's Singleton Weight Sharing is "architecture-default expansion." The two's quality consequences should be consistent: after $k > 2$ quality flattens.

This gives an actionable prediction: **running voting/ensembles on a Warp-Cortex-like singleton-weight architecture, the curve of quality with N should approach P9's $\$1/k$ saturation, not continuously fall.** If someone uses a singleton-weight-sharing architecture to claim "1000 Agents' ensemble quality continuously improves," this paper predicts its gains will be narrowed in third-party reproduction or attributed to the strongest single member (i.e. α itself).

4.2 The Topological Synapse Retains "Shape," Not "Independence"

Worth supplementing is the subtlety of the Topological Synapse. It retains the KV-cache point cloud's "persistent-homology features" through TDA landmarking—a good kind of **information compression**, making each Agent's context more memory-efficient. But compression retains that Agent's own point cloud's geometric shape and does not make different Agents' point clouds mutually independent. That is, the topological synapse optimizes "whether a single Agent's own memory saves memory," not "whether information between Agents is independent." It serves the memory wall, not the quality wall.

4.3 What Is the Real Quality Countermeasure

Per P20's conclusion, to break through the quality wall, what is needed is not "letting 1000 same-source Agents share one weight" but "letting a few Agents carry truly independent information sources"—different models, different tools, different data. Warp-Cortex solves "affording same-source Agents," P9/P20 point out "affordable does not equal well-raised." Combining the two is the complete engineering

picture: **use singleton weights to press down \$N\$'s memory cost, then use heterogeneous information sources to raise \$D\$.**

4.4 A Key Distinction from MoE Routing

Here is an easily confused contrast worth clarifying. Switch Transformer (Fedus et al., 2021, arXiv: 2101.03961) uses top-1 sparse routing to bring total parameters to about 1.6T, but it and Warp-Cortex's singleton weight sharing go in opposite directions on "diversity." MoE is **stuffing multiple experts into one model, routing by token condition**—different tokens sent to different experts, and experts indeed carry different subspace knowledge; Warp-Cortex is **multiple Agents sharing one weight, differing only on KV-cache**. The former is "intra-model heterogenization," the latter "inter-Agent homogenization." So MoE's routing balance (whether experts collapse) is its quality problem, while Warp-Cortex's quality problem defaults from architecture to "homogenized-copy saturation." This distinction reminds us: **"large parameter count" and "independent information source" are not the same thing**—MoE's total parameters are large, but if routing collapses to a few experts, the equivalent independent information sources also shrink.

4.5 Is Referential Injection a Memory or Coordination Mechanism

Warp-Cortex's Referential Injection allows asynchronous sub-Agents to influence main generation without interrupting the main stream. This mechanism is often misread as a "coordination breakthrough," but from this paper's tripartition it looks more like a **memory and scheduling mechanism**: it solves "how asynchronous sub-Agents' results are injected without blocking main inference," i.e. scheduling and context management; it does not solve fan-in, consensus, or fault propagation among multiple Agents. Reading it as "the coordination wall is also torn down" crosses over P1's work. The correct reading is: Referential Injection makes parallel asynchronous execution under the singleton-weight architecture more memory-efficient, still within the memory wall.

4.6 Corollary for UDOS Architecture: Layered Hybrid, Not Full Homogenization

Landing the three-wall division on UDOS's architecture decisions yields an executable combination scheme. UDOS need not choose either-or—neither blindly pursuing "million same-source copies" nor blindly opposing shared weights. The reasonable layering is: **bottom-layer execution copies use singleton weight sharing (taking the memory-wall dividend, responsible for throughput), top-layer judgment members use heterogeneous independent information sources (taking the quality-wall dividend, responsible for decisions)**. That is, Warp-Cortex-style shared weights suit running large numbers of "sampling-style, approximately repeated" execution tasks (such as batch candidate generation, parallel scenario simulation), while the independent information sources P9 emphasizes should be retained at the few key positions truly making rulings (such as the QA committee, completion decisions).

This combination is in spirit fully consistent with P1's layered hybrid tree: the bottom Swarm can be homogeneous copies (cheap, parallel), the middle Handoff heterogeneous specialists, the top Orchestrator independent-source judgment. The memory wall is torn down at the bottom by shared weights, the quality wall held at the top by heterogeneous sources. Making the whole system singleton-weight-shared equals handing the quality wall back.

Summarizing this corollary in one sentence: **the correct position of Warp-Cortex-like architectures is "scaling of the execution layer," not "qualitization of the decision layer"**. If what you need is hundreds or thousands of approximately repeated executions (sampling, simulation, batch generation),

singleton weight sharing is the correct memory-saving choice; if what you need is more accurate judgment, the correct investment remains replacing a few key members with truly independent information sources. Confusing these two layers is the most cost-effective self-comfort in the 2026 "million-Agent" narrative.

5 Dross, Failure Boundaries, and Falsifiable Conditions

5.1 Dross and circulated exaggeration

1. **Taking the title's "million" as measurement.** Warp-Cortex actually measured only about 100 Agents/2.2GB, the thousand level is theoretical extrapolation, the million title narrative and the paper self-notes "before compute latency becomes the bottleneck." Writing it as "already running million Agents on consumer hardware" is taking extrapolation as measurement.
2. **Reading "memory scaling" as "capability scaling."** Singleton weight sharing is homogenization, which enlarges $\$N\$$ while pinning $\$D\$$. This is the core equivocation this paper corrects.
3. **Ignoring the coordination wall.** Warp-Cortex only discusses memory, not communication fan-in, consensus cost, fault propagation among 100 Agents. Reading "memory fits 1000" as "1000 can coordinate" crosses over P1's work.

5.2 Failure boundaries

- This paper neither denies Warp-Cortex's memory contribution nor the value of singleton weight sharing in scenarios "needing many parallel but mutually approximate Agents" (such as batch sampling the same policy); this paper only draws clear its applicability boundary to **ensemble quality**.
- If KV-cache differentiation is large enough that different Agents actually reach independent information pathways, then singleton weights + differentiated context may also partially break through $\$b\to0\$$; this needs experimental verification, and this paper conservatively treats it per the architecture-default homogenization.

5.3 Falsifiable conditions

- **C1:** if on a Warp-Cortex-like singleton-weight architecture, ensemble quality continuously and significantly falls with concurrent Agent count $\$N\$$ (not flattening, gains not explained by the strongest single member), then the "singleton weight = homogenization = saturation" corollary is weakened.
- **C2:** if measurement (not extrapolation) stably runs above the thousand-level concurrency on consumer hardware with quality not degrading, then this paper's "measured only the hundred level" caliber needs upward revision.

6 Hard Predictions (2026–2031)

- **F1:** over the next five years, for systems taking "singleton weight sharing + large-scale concurrency" as the selling point, the claim of quality rising with $\$N\$$ will mostly be narrowed in third-party reproduction to the strongest single member level; the true quality increment still comes from heterogeneous information sources, not concurrency count.
- **F2:** the first public system to simultaneously deliver "memory-wall breakthrough" and "quality-wall breakthrough" will clearly distinguish "shared-weight execution copies" and "independent-source judgment members"—the former responsible for throughput, the latter for quality.

7 Conclusion

Warp-Cortex tore down the memory wall: singleton weight sharing presses the weight part to $O(1)$, the topological synapse presses context to $O(N \cdot k)$, the paper reports one 4090 running 100 Agents. This is a real contribution. But the singleton weight sharing it uses is architecturally what P9 calls same-source replication—it lets N reach hundreds/theoretical thousands while pinning D at 1. Memory can scale, coordination can scale, quality can scale are three things: Warp-Cortex handles the first, P1 the second, P9/P20 the third.

For engineering, this means before claiming a "million-Agent system" one must first answer the three walls: does memory fit (Warp-Cortex answers)? Do messages explode (P1 answers)? Does adding Agents raise quality (P9 answers)? Only when the three answers are "yes" is the million level worth discussing. For research, it gives a falsifiable prediction (C1): the quality curve on a singleton-weight architecture should saturate.

Tearing down the memory wall does not equal tearing down the quality wall. Scale can go up in quantity does not equal capability can go up a tier.

References

Background and classics

1. Ruiz Williams, J. L. (2026). Warp-Cortex: An Asynchronous, Memory-Efficient Architecture for Million-Agent Cognitive Scaling on Consumer Hardware. arXiv:2601.01298.
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UDOS paper volume (this series, evidence grades marked in text)

1. UDOS v7.5.0: P9 "The Quality Saturation Law of Agent Legions and the Failure Boundary of Homogenized Expansion", P1 "Million-Scale Extrapolation of Coordination Complexity of Layered Hybrid Topologies".
 2. UDOS v7.6.0: P20 "Heterogeneity Dividend and Homogeneity Saturation", P21 "End-to-End Evidence for Million-Scale Coordination and Its Extrapolation Boundary".
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Evidence Discipline and Reproduction Notes

- This paper is a theory/architecture-analysis paper with no new experiments; all UDOS numbers return to P9/P1 and [reports7/](#), evidence grades per each paper's labeling (cpu-proto / verified).
- Warp-Cortex's "100 Agents/2.2GB" is paper-reported, "thousand/million" the paper's self-stated theoretical extrapolation, both labeled "paper-reported • not independently rechecked"; the mechanisms of Singleton Weight Sharing and Topological Synapse are verified facts.
- This paper's predictions (F1–F2) and falsification conditions (C1–C2) can be post-hoc scored with public evidence during 2026–2031.